

Weichen Huang

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RESEARCH INTEREST

Mechanistic interpretability of deep neural networks; representation geometry in language and vision models; computational links between artificial and biological intelligence.

RESEARCH EXPERIENCE

- **Language Intelligence Lab @ Georgia Tech** Atlanta, GA
Language Network Research / with PI: Dr. Anna Ivanova Oct 2025 – Present
 - Probed hidden activations using finetuned linear probes to test which representational subspaces best predict neural responses
 - Explored prompt recovery from hidden activations as a diagnostic tool for semantic compression in language models.
 - Implemented representational similarity analysis between LLM hidden states and voxel-wise fMRI responses to study layer-wise semantic alignment
- **DiCarlo Lab @ MIT (RSI 2024)** Cambridge, MA
Neuromorphic computational vision / with Yudi Xie and PI: Dr. James J. DiCarlo Jun 2024 – Oct 2024
 - Investigated and implemented representation similarity methods like CKA and RSA.
 - Formulated and tested hypotheses relating representation geometry to architectural inductive biases using CKA/RSA analyses.
 - **Published to ICLR 2025: Vision Models Trained to Estimate Spatial Latents Learn Ventral-Stream-Aligned Representations.**
 - **Featured in MIT News:**
<https://news.mit.edu/2025/visual-pathway-brain-may-do-more-than-recognize-objects-0415>
- **ML in Medical Imaging Lab @ UCD** Dublin, Ireland
Deep learning for medical imaging and multimodal diagnosis / with PI: Dr. Kathleen Curran Feb 2023 – Jan 2025
 - Studied spurious correlations in chest X-ray (CXR) interpretation models; applied reinforcement learning and self-/semi-supervised training to improve robustness.
 - Developed MRI-PET fusion technique with dynamic image compression for Alzheimer's disease diagnosis; evaluated on ADNI with improvements over prior baselines.
 - Built multimodal contrastive learning framework combining MRI images with tabular clinical features via tabular attention; improved Alzheimer's prediction accuracy to 83.8%+.
 - Proposed graph-based contrastive learning framework enabling learning from an arbitrary number of modalities, improving generalizability.
 - Research published in IMVIP 2023, ICCV 2023; NeurIPS 2024.

CONFERENCE PUBLICATIONS

- Xie, Y., **Huang, W.**, Alter, E., Schwartz, J., Tenenbaum, J., DiCarlo, J. (2025). *Vision Models Trained to Estimate Spatial Latents Learned Similar Ventral-stream-aligned Representations*. Contribution: model analysis, representation similarity experiments, hypothesis testing. OpenReview
- **Huang, W.** (2023). *Multimodal Contrastive Learning and Tabular Attention for Automated Alzheimer's Disease Prediction*. ICCV 2023, pp. 2465–2474. arXiv:2308.15469.
- **Huang, W.**, Curran, K. (2023). *Spurious Correlation Mitigation in CXR Images via Reinforcement Learning and Self-Supervision*. IMVIP 2023. Zenodo.
- **Huang, W.** (2022). *Feature Scaling and Attention Convolutions for Extreme Precipitation Prediction*. Fragile Earth 2022 (ACM SIGKDD workshop). PDF.

PREPRINTS & OTHER PUBLICATIONS

- **Huang, W.** (2024). *Multimodal Representation Learning using Adaptive Graph Construction*. NeurIPS 2024 (High School Projects track; top 4/335). arXiv:2410.06395.
- Chang, Z., **Huang, W.**, Curran, K. (2023). *Improved Multimodal Diagnosis of Alzheimer's Disease through MRI and PET Image Fusion*. RSNA 2023.

SELECTED PROJECTS & DEMOS

- **Vision Model Research (ICLR25)** Vision Models Trained to Estimate Spatial Latents Learned Similar Ventral-stream-aligned Representations. <https://github.com/YudiXie/multitask-vision>
- **Multimodal Contrastive Learning (ICCV23)**. Multimodal Contrastive Learning and Tabular Attention for Automated Alzheimer's Disease Prediction Github
- **CXR Predict (demo)**. Self-supervised cross-modal representation learning. Hugging Face Space.
- **NYT Connections Solver** Solving NYT Connections word game using language model entropy. Github
- **Extreme weather prediction using attention-augmented convolutions**. Project page: weichen-huang.github.io/projects/saconvnets.
- **EfficientNet COVID-19 detection from lung X-ray imaging**. Project page: weichen-huang.github.io/projects/covidEfficientnet.

SKILLS

- **ML/DL**: PyTorch, PyTorch Lightning, Hugging Face, self-supervised & contrastive learning
- **Interpretability**: linear probes, sparse autoencoders, RSA, representation geometry
- **Neuroscience**: fMRI encoding models, voxel-wise regression, multimodal alignment
- **Math**: linear algebra, probability, optimization

EDUCATION

- **Georgia Institute of Technology** Atlanta, GA
B.S. in Computer Science (Class of 2029); GPA: 4.0/4.0 2025 – Present
 - Selected coursework: Theoretical Linear Algebra (MATH 3406), Probability & Statistics (MATH 3215), Applied Combinatorics (MATH 3012), General Psychology (PSYC 1101).
 - Competitive programming: Georgia Tech ICPC Regional (2025).
- **St Andrew's College** Dublin, Ireland
International Baccalaureate; 43/45; SAT 1580/1600; DELF (French) B1 Graduated Jun 2025
 - HL: Mathematics Analysis & Approaches (7), Physics (7), Chemistry (7).

AWARDS & HONORS

- **Competitive Programming**: IOI Irish Team (2025), USACO Gold (since 2023);
- **Research**: NeurIPS high school track top publication (2024).